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United States Patent	12408419
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lin; Meng-Han et al.

Semiconductor device and method for forming capacitor structure

Abstract

A semiconductor device includes a first device over a substrate, wherein the first device includes a gate stack including a gate electrode material; a source/drain region in the substrate adjacent the gate stack; a first isolation region surrounding the gate stack; a gate contact over and contacting the gate stack, wherein the gate contact includes a gate contact material; and a second isolation region surrounding the gate contact; and a second device over the substrate, wherein the second device includes a first parallel capacitor including first electrodes, wherein the first electrodes include the gate electrode material, wherein the first isolation region separates the first electrodes; and a second parallel capacitor over the first parallel capacitor, wherein the second parallel capacitor includes second electrodes connected to the first electrodes, wherein the second electrodes include the gate contact material, wherein adjacent second electrodes are separated by the second isolation region.

Inventors: Lin; Meng-Han (Hsinchu, TW), Yeong; Sai-Hooi (Zhubei, TW), Chui; Chi On (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)

Family ID: 83574896

Assignee: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)

Appl. No.: 17/647046

Filed: January 05, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220392889 A1	Dec. 08, 2022

Related U.S. Application Data

us-provisional-application US 63196971 20210604

Publication Classification

Int. Cl.: **H10D84/80** (20250101); **H10B12/00** (20230101); **H10D1/68** (20250101); **H10D30/01** (20250101); **H10D30/62** (20250101); **H10D30/69** (20250101); **H10D62/822** (20250101); **H10D64/01** (20250101); **H10D84/01** (20250101); **H10D84/03** (20250101); **H10D84/83** (20250101)

U.S. Cl.:

CPC **H10D84/811** (20250101); **H10D1/692** (20250101); **H10D30/024** (20250101); **H10D30/6211** (20250101); **H10D64/017** (20250101);

Field of Classification Search

CPC: H01L (27/0629); H01L (28/60); H01L (29/66545); H01L (29/66795); H01L (29/7851); H01L (21/823431); H01L (27/0886); H01L (29/165); H01L (29/6656); H01L (29/7848); H10B (12/36); H10B (12/03); H10B (12/056)

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Primary Examiner: Assouman; Herve-Louis Y

Attorney, Agent or Firm: Slater Matsil, LLP

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of U.S. Provisional Application No. 63/196,971, filed on Jun. 4, 2021, which application is hereby incorporated herein by reference.

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

(2) The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates an example of a FinFET structure in a three-dimensional view, in accordance with some embodiments.

(3) FIGS. 2, 3, 4, 5, 6, 7, 8, 9A, 9B, 9C, 9D, 10A, 10B, 10C, 10D, 11A, 11B, 11C, 11D, 11E, 11F, 12A, 12B, 12C, 12D, 13, 14A, 14B, 14C, 14D, 15, 16A, 16B, 16C, 16D, 17, 18A, 18B, 18C, 18D, 18E, 19A, 19B, 19C, 19D, 20, 21A, 21B, 21C, 21D, 22A, 22B, 23, 24A, 24B, 25, 26A, and 26B are various views of intermediate stages in the manufacturing of a capacitor structure and a FinFET structure, in accordance with some embodiments.

(4) FIG. 27 illustrates an example of a capacitor structure in a three-dimensional view, in accordance with some embodiments.

(5) FIG. 28 illustrates an example of an interdigitated capacitor structure in a three-dimensional view, in accordance with some embodiments.

(6) FIGS. 29 and 30 are plan views of intermediate stages in the manufacturing of an interdigitated capacitor structure, in accordance with some embodiments.

(7) FIGS. 31A and 31B are cross-sectional views of intermediate stages in the manufacturing of a capacitor structure and a planar transistor structure, in accordance with some embodiments.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in

itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) Various embodiments describe processes for forming a capacitor structure. The capacitor structure may be, for example, a Metal-Insulator-Metal (MIM) capacitor, a Metal-Oxide-Metal (MOM) capacitor, a Finger Metal-Oxide-Metal (FMOM) capacitor, or the like. The capacitor structure described herein includes bottom electrodes formed simultaneously with the gate electrode of a transistor and formed using the same processes that form the gate electrode of the transistor. In this manner, additional electrodes of a capacitor structure may be formed to increase capacitance without additional processing steps. Contacts to the bottom electrodes may be formed that also provide additional capacitance. The electrode contacts may be formed simultaneously with the gate contacts and/or source/drain contacts of the transistor, and formed using the same processes that form the gate contacts and/or source/drain contacts of the transistor. In this manner, the capacitance of a capacitor structure may be increased without additional processing steps. The techniques described herein may include processes suitable for forming n-type transistors and/or p-type transistors, and may be applicable to different kinds of transistors.

(11) FIGS. 1 through 27 are various views of intermediate stages in the manufacturing of a capacitor structure 120 (see FIGS. 26A-26B) and a fin field-effect transistor (FinFET) structure 119 (see FIGS. 21C-21D), in accordance with some embodiments. In some embodiments, the capacitor structure 120 and the FinFET structure 119 are formed simultaneously on the same substrate 50, and are formed using at least some of the same process steps. In some embodiments, the capacitor structure 120 and the FinFET structure 119 are formed entirely using the same process steps. In this manner, a capacitor structure 120 may be formed without the use of additional process steps or masks, which can reduce the manufacturing cost of a device.

(12) Some embodiments discussed herein are discussed in the context of transistors (e.g., FinFETs or planar FETs) formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in other devices, such as nanostructure (e.g., nanosheet, nanowire, gate-all-around, or the like) field effect transistors (NSFETs), or the like.

(13) Turning first to FIG. 1, an example of a FinFET in a three-dimensional view is shown, in accordance with some embodiments. The FinFET shown in FIG. 1 is an example structured used as a reference for the subsequent discussion of the process steps used in the manufacturing of the FinFET structure 119 (see FIGS. 21C-21D). The FinFET shown in FIG. 1 comprises a fin 52 on a substrate 50 (e.g., a semiconductor substrate). Isolation regions 56 are disposed in the substrate 50, and the fin 52 protrudes above and from between neighboring isolation regions 56 (e.g., Shallow Trench Isolation (STI) regions 56). Although the isolation regions 56 are described/illustrated as being separate from the substrate 50, as used herein the term “substrate” may be used to refer to just the semiconductor substrate or a semiconductor substrate inclusive of isolation regions. Additionally, although the fin 52 is illustrated as a single, continuous material as the substrate 50, the fin 52 and/or the substrate 50 may comprise a single material or a plurality of materials. In this context, the fin 52 refers to the portion extending between the neighboring isolation regions 56.

(14) A gate dielectric layer 92 is along sidewalls and over a top surface of the fin 52, and gate electrode layers 94 are over the gate dielectric layer 92. Source/drain regions 82 are disposed in opposite sides of the fin 52 with respect to the gate dielectric layer 92 and gate electrode layers 94. FIG. 1 further illustrates reference cross-sections that are used in later figures. Cross-section C-C is

along a longitudinal axis of the gate electrode layers **94** and in a direction, for example, perpendicular to the direction of current flow between the source/drain regions **82** of the FinFET. Cross-section D-D is perpendicular to cross-section C-C and is along a longitudinal axis of the fin **52** and in a direction of, for example, a current flow between the source/drain regions **82** of the FinFET. Cross-section E-E is parallel to cross-section C-C and extends through a source/drain region **82** of the FinFET. Subsequent figures refer to these reference cross-sections for clarity.

(15) FIGS. **2** through **7** are various views of intermediate stages in the manufacturing of a capacitor structure **120** and a FinFET structure **119**, in accordance with some embodiments. FIGS. **2**, **3**, **4**, **6**, and **7** illustrate the transistor region **50X** along the reference cross-section C-C illustrated in FIG. **1**, except for multiple fins/FinFETs. FIG. **5** illustrates a plan view of the capacitor region **50C** and the transistor region **50X**.

(16) In FIG. **2**, a substrate **50** is provided. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; the like; or combinations thereof.

(17) The substrate **50** is shown having a capacitor region **50C** and a transistor region **50X**, in accordance with some embodiments. The capacitor region **50C** is a region in which one or more capacitor structures **120** are formed, and the transistor region **50X** is a region in which one or more transistors such as FinFET structures **119** are formed. A single capacitor region **50C** and a single transistor region **50X** are shown in the figures, but a substrate may have any suitable number of capacitor regions **50C** or transistor regions **50X** of any suitable sizes. Other types of devices or structures than capacitors may be formed in the capacitor region **50C**, and other types of devices or structures than transistors may be formed in the transistor region **50X**. The capacitor region **50C** may be physically separated from the transistor region **50X** (as illustrated by divider **51**), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the capacitor region **50C** and the transistor region **50X**. Further, the transistor region **50X** may comprise an n-type region for forming n-type devices, a p-type region for forming p-type devices, or both an n-type region and a p-type region. The n-type region and the p-type region of the transistor region **50X** may be referred to herein as “type regions.”

(18) In FIG. **3**, fins **52** are formed in the substrate **50** in the transistor region **50X**, in accordance with some embodiments. The fins **52** are semiconductor strips. In some embodiments, the fins **52** may be formed in the substrate **50** by etching trenches in the substrate **50**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic. In some embodiments, the etch also etches the substrate **50** in the capacitor region **50C**, as shown in FIG. **3**.

(19) The fins **52** may be patterned by any suitable method. For example, the fins **52** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using

a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins. In some embodiments, the mask (or other layer) may remain on the fins **52**.

(20) In FIG. **4**, an insulation material **54** is formed over the substrate **50** and between neighboring fins **52** in the transistor region **50X**, in accordance with some embodiments. The insulation material **54** may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material **54** is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material **54** is formed such that excess insulation material **54** covers the fins **52**. Although the insulation material **54** is illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not shown) may first be formed along a surface of the substrate **50** and the fins **52**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

(21) Still referring to FIG. **4**, a removal process may be applied to the insulation material **54** to remove excess insulation material **54** over the fins **52**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the fins **52** such that top surfaces of the fins **52** and the insulation material **54** are level after the planarization process is complete. In embodiments in which a mask remains on the fins **52**, the planarization process may expose the mask or remove the mask such that top surfaces of the mask or the fins **52**, respectively, and the insulation material **54** are level after the planarization process is complete. As shown in FIG. **4**, the top surfaces of the insulation material **54** in the capacitor region **50C** and in the transistor region **50X** may be approximately level after performing the planarization process.

(22) In FIGS. **5** and **6**, the insulation material **54** is recessed to form Shallow Trench Isolation (STI) regions **56**, in accordance with some embodiments. FIG. **5** illustrates a plan view of the capacitor region **50C** and the transistor region **50X**. FIG. **6** illustrates a cross-sectional view through the reference cross-section C-C shown in FIG. **5**. The insulation material **54** is recessed such that upper portions of fins **52** in the transistor region **50X** protrude from between neighboring STI regions **56**. Further, the top surfaces of the STI regions **56** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **56** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **56** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material **54** (e.g., etches the material of the insulation material **54** at a faster rate than the material of the fins **52**). For example, an oxide removal process using dilute hydrofluoric acid (dHF) may be used, though other processes are possible.

(23) The process described with respect to FIGS. **2** through **6** is just one example of how the fins **52** in the transistor region **50X** may be formed. In some embodiments, the fins **52** may be formed by an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer to expose the underlying substrate **50**. Homoepitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins. Additionally, in some embodiments, heteroepitaxial structures can be used for the fins **52**. For example, the fins **52** in FIG. **6** can be recessed, and a material different from the fins **52** may be epitaxially grown over the recessed fins **52**. In such embodiments, the fins **52** comprise the recessed material as well as the epitaxially grown material disposed over the recessed material. In an even further embodiment, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer. Heteroepitaxial structures can then be

epitaxially grown in the trenches using a material different from the substrate **50**, and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form the fins **52**. In some embodiments where homoepitaxial or heteroepitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and subsequent implantations although in situ and implantation doping may be used together.

(24) Still further, it may be advantageous to epitaxially grow a material in an n-type region of the transistor region **50X** that is different from the material in a p-type region of the transistor region **50X**. In various embodiments, upper portions of the fins **52** may be formed from silicon germanium ($\text{Si.sub.xGe.sub.1-x}$, where x can be in the range of 0 to 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, indium arsenide, aluminum arsenide, gallium arsenide, indium phosphide, gallium nitride, indium gallium arsenide, indium aluminum arsenide, gallium antimonide, aluminum antimonide, aluminum phosphide, gallium phosphide, and the like.

(25) Further in FIG. **6**, appropriate wells (not shown) may be formed in the fins **52** and/or the substrate **50**. In some embodiments, a P well may be formed in the n-type region of the transistor region **50X**, and an N well may be formed in the p-type region of the transistor region **50X**. In some embodiments, a P well or an N well is formed in both the n-type region and the p-type region of the transistor region **50X**. A P well and/or an N well may be formed in the capacitor region **50C**, in some embodiments.

(26) In the embodiments with different well types, the different implant steps for the n-type region and the p-type region of the transistor region **50X** may be achieved using a photoresist and/or other masks (not shown). For example, a photoresist may be formed over the fins **52** and the STI regions **56** in the n-type region. The photoresist is patterned to expose the p-type region of the transistor region **50X**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, the n-type impurity implant is performed in the p-type region, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the n-type region. The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration of equal to or less than $10.\text{sup.18 cm.sup.-3}$, such as between about $10.\text{sup.16 cm.sup.-3}$ and about $10.\text{sup.18 cm.sup.-3}$. After the implant, the photoresist is removed, such as by an acceptable ashing process.

(27) Following the implanting of the p-type region, a photoresist is formed over the fins **52** and the STI regions **56** in the p-type region. The photoresist is patterned to expose the n-type region of the transistor region **50X**. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implant may be performed in the n-type region, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the p-type region. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration of equal to or less than $10.\text{sup.18 cm.sup.-3}$, such as between about $10.\text{sup.16 cm.sup.-3}$ and about $10.\text{sup.18 cm.sup.-3}$. After the implant, the photoresist may be removed, such as by an acceptable ashing process.

(28) After the implants of the n-type region and the p-type region, an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

(29) In FIG. **7**, a dummy dielectric layer **60** is formed on the fins **52** in the transistor region **50X**, in accordance with some embodiments. FIG. **7** is illustrated along the same cross-section C-C as FIG. **6**. The dummy dielectric layer **60** may be, for example, silicon oxide, silicon nitride, a combination

thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **62** is formed over the capacitor region **50C** and the transistor region **50X**, including over the dummy dielectric layer **60**. A mask layer **64** is then formed over the dummy gate layer **62**, in some embodiments. The dummy gate layer **62** may be deposited and then planarized, such as by a CMP or the like. The mask layer **64** may then be deposited over the dummy gate layer **62**. The dummy gate layer **62** may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **62** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. The dummy gate layer **62** may be made of other materials that have a high etching selectivity from the etching of isolation regions, e.g., the STI regions **56** and/or the dummy dielectric layer **60**. The mask layer **64** may include one or more layers of, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer **62** and a single mask layer **64** are formed across the capacitor region **50C** and the transistor region **50X**. It is noted that the dummy dielectric layer **60** is shown covering only the fins **52** for illustrative purposes only. In some embodiments, the dummy dielectric layer **60** may be deposited such that the dummy dielectric layer **60** covers the STI regions **56**, extending over the STI regions **56** and between the dummy gate layer **62** and the STI regions **56**.

(30) FIGS. **8** through **27** illustrate various additional steps in the manufacturing of embodiment devices. The structures illustrated in the transistor region **50X** may be applicable to both an n-type region and a p-type region. Differences (if any) in the structures of an n-type region and a p-type region of the transistor region **50X** are described in the text accompanying each figure.

(31) FIGS. **8** through **21D** illustrate plan views or cross-sectional views of features in the capacitor region **50C** and/or the transistor region **50X**. FIGS. **8**, **13**, **15**, and **17** are plan views of the capacitor region **50C** and the transistor region **50X**, with cross-sections A-A and B-B indicated in the capacitor region **50C** and cross-sections C-C and D-D indicated in the transistor region **50X**. The cross-sections C-C and D-D correspond to the cross-sections C-C and D-D shown in FIG. **1**. Cross-section A-A is parallel to cross-section C-C, cross-section B-B is parallel to cross-section D-D, and cross-sections A-A and C-C are perpendicular to cross-sections B-B and D-D. The cross-section A-A may or may not be aligned with cross-section C-C, and the cross-section B-B may or may not be aligned with cross-section D-D.

(32) FIGS. **9A**, **10A**, **11A**, **12A**, **14A**, **16A**, **18A**, **19A**, and **21A** are cross-sectional views of the capacitor region **50C** along cross-section A-A (e.g., as shown in FIG. **8**). FIGS. **9B**, **10B**, **11B**, **12B**, **14B**, **16B**, **18B**, **19B**, and **21B** are cross-sectional views of the capacitor region **50C** along cross-section B-B. FIGS. **9C**, **10C**, **11C**, **12C**, **14C**, **16C**, **18C**, and **19C** are cross-sectional views of the transistor region **50X** along cross-section C-C. FIGS. **9D**, **10D**, **11D**, **12D**, **14D**, **16D**, **18D**, and **19D** are cross-sectional views of the transistor region **50X** along cross-section D-D.

(33) In FIGS. **8** and **9A-9D**, the mask layer **64** is patterned to form masks **74**, and the dummy gate layer **62** is patterned to form dummy gates **72**, in accordance with some embodiments. As stated previously, FIG. **8** illustrates a plan view and FIGS. **9A-9D** illustrate corresponding cross-sectional views. The mask layer **64** (see FIG. **7**) may be patterned using acceptable photolithography and etching techniques to form masks **74**. The pattern of the masks **74** then may be transferred to the dummy gate layer **62** to form dummy gates **72**. In some embodiments (not illustrated), the pattern of the masks **74** may also be transferred to the dummy dielectric layer **60** by an acceptable etching technique to form dummy gates **72**. The dummy gates **72** extend over the STI regions **56** in the capacitor region **50C**, and cover respective channel regions **58** of the fins **52** in the transistor region **50X**. The pattern of the masks **74** may be used to physically separate each of the dummy gates **72** from adjacent dummy gates **72**.

(34) In some embodiments, the dummy gates **72** in the capacitor region **50C** are patterned to form two separated structures corresponding to two bottom electrodes of a capacitor structure. For

example, the dummy gates **72** shown in FIGS. **8** and **9A-9B** have been patterned to form a first dummy bottom electrode **95A'** and a second dummy bottom electrode **95B'**. The first dummy bottom electrode **95A'** and the second dummy bottom electrode **95B'** are subsequently processed to form a first bottom electrode **95A** and a second bottom electrode **95B** (see FIGS. **17** and **18A-18B**) of a capacitor structure **120** (see FIGS. **26A-26B** and **27**).

(35) In some embodiments, the dummy gates **72** in the transistor region **50X** have a lengthwise direction that is substantially perpendicular to the lengthwise direction of respective fins **52**, as shown in FIGS. **8** and **9C-9D**. FIGS. **8** and **9A-9B** also illustrate the dummy gates **72** in the capacitor region **50C** (e.g., the dummy bottom electrodes **95A'** and **95B'**) as having a lengthwise direction that is parallel to the lengthwise direction of the dummy gates **72** in the transistor region **50X**. In other embodiments, the dummy gates **72** in the capacitor region **50C** have a lengthwise direction that is perpendicular to the lengthwise direction of the dummy gates **72** in the transistor region **50X**. In some embodiments, the dummy gates **72** in the capacitor region **50C** may comprise both parallel portions and perpendicular portions, or may comprise portions that are neither parallel nor perpendicular (e.g., rounded, oblique, irregular, etc.) to the lengthwise direction of the dummy gates **72** in the transistor region **50X**.

(36) The first dummy bottom electrode **95A'** and the second dummy bottom electrode **95B'** shown in FIGS. **8** and **9A-9B** are illustrated examples, and in other embodiments the dummy bottom electrodes **95A'** and **95B'** may have different dimensions, different shapes, or a different arrangement than shown. As non-limiting examples, in other embodiments, the dummy bottom electrodes **95A'** and **95B'** may comprise more than two electrodes, “L-shaped” electrodes, or an arrangement of interdigitated “fingers.” An example of an interdigitated capacitor structure **130** is described below for FIGS. **28-30**, though other shapes or arrangements are possible.

(37) In FIGS. **10A-10D**, gate seal spacers **80** and gate spacers **86** are formed in the capacitor region **50C** and the transistor region **50X**, in accordance with some embodiments. The gate seal spacers **80** are formed on exposed surfaces of the dummy gates **72**, the masks **74**, and/or the fins **52**. The gate seal spacers **80** may be formed, for example, by performing a thermal oxidation or a deposition followed by an anisotropic etch. The gate seal spacers **80** may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like.

(38) After the formation of the gate seal spacers **80**, implants for lightly doped source/drain (LDD) regions (not explicitly illustrated) may be performed in the transistor region **50X**. In the embodiments in which the transistor region **50X** includes different device types, the implants may be performed using techniques similar to the implants discussed above for FIG. **6**. For example, a mask, such as a photoresist, may be formed over the n-type region of the transistor region **50X**, while exposing the p-type region of the transistor region **50X**, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins **52** in the p-type region. The mask may then be removed. Subsequently, a mask, such as a photoresist, may be formed over the p-type region while exposing the n-type region, and appropriate type impurities (e.g., n-type) may be implanted into the exposed fins **52** in the n-type region. The mask may then be removed. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a concentration of impurities of from about 10^{15} cm.⁻³ to about 10^{19} cm.⁻³. An anneal may be used to repair implant damage and to activate the implanted impurities. In some embodiments, impurities are also implanted in the capacitor region **50C**.

(39) Still referring to FIGS. **10A-10D**, gate spacers **86** are formed on the gate seal spacers **80** along sidewalls of the dummy gates **72** and the masks **74**, in accordance with some embodiments. The gate spacers **86** may be formed, for example, by conformally depositing an insulating material and subsequently anisotropically etching the insulating material. The insulating material of the gate spacers **86** may be silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, a combination thereof, or the like. The gate spacers **86** may be formed from one layer of insulating

material or from multiple layers of various insulating materials.

(40) It is noted that the above disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized or different sequence of steps may be utilized (e.g., the gate seal spacers **80** may not be etched prior to forming the gate spacers **86** which may yield “L-shaped” gate seal spacers, spacers or layers thereof may be formed and removed, and/or the like). Furthermore, the n-type and p-type devices may be formed using a different structures and steps. For example, LDD regions for n-type devices may be formed prior to forming the gate seal spacers **80** while the LDD regions for p-type devices may be formed after forming the gate seal spacers **80**.

(41) In FIGS. **11A-11D**, epitaxial source/drain regions **82** are formed in the fins **52** in the transistor region **50X**, in accordance with some embodiments. The epitaxial source/drain regions **82** are formed in the fins **52** such that each dummy gate **72** in the transistor region **50X** is disposed between respective neighboring pairs of the epitaxial source/drain regions **82**. In some embodiments, the epitaxial source/drain regions **82** may extend into, and may also penetrate through, the fins **52**. In some embodiments, the gate spacers **86** are used to separate the epitaxial source/drain regions **82** from the dummy gates **72** by an appropriate lateral distance so that the epitaxial source/drain regions **82** do not short out subsequently formed gates of the resulting FinFETs. A material of the epitaxial source/drain regions **82** may be selected to exert stress in the respective channel regions **58**, thereby improving performance.

(42) The epitaxial source/drain regions **82** in the n-type region of the transistor region **50X** may be formed, for example, by masking the p-type region of the transistor region **50X** and etching source/drain regions of the fins **52** in the n-type region to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the n-type region are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include any acceptable material, such as appropriate for n-type FinFETs. For example, if the fin **52** is silicon, the epitaxial source/drain regions **82** in the n-type region may include materials exerting a tensile strain in the channel region **58**, such as silicon, silicon carbide, phosphorous-doped silicon carbide, silicon phosphide, or the like. The epitaxial source/drain regions **82** in the n-type region may have surfaces raised from respective surfaces of the fins **52** and may have facets.

(43) The epitaxial source/drain regions **82** in the p-type region may be formed by masking the n-type region and etching source/drain regions of the fins **52** in the p-type region to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the p-type region are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include any acceptable material, such as appropriate for p-type FinFETs. For example, if the fin **52** is silicon, the epitaxial source/drain regions **82** in the p-type region may comprise materials exerting a compressive strain in the channel region **58**, such as silicon germanium, boron-doped silicon germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions **82** in the p-type region may have surfaces raised from respective surfaces of the fins **52** and may have facets.

(44) The epitaxial source/drain regions **82** and/or the fins **52** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an anneal. The source/drain regions may have an impurity concentration of between about 10^{19} cm⁻³ and about 10^{21} cm⁻³. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions **82** may be in situ doped during growth.

(45) As a result of the epitaxy processes used to form the epitaxial source/drain regions **82** in the n-type region and/or the p-type region of the transistor region **50X**, upper surfaces of the epitaxial source/drain regions **82** have facets which expand laterally outward beyond sidewalls of the fins **52**. This is shown in FIGS. **11E** and **11F**, which illustrate example cross-sectional views along the cross-section E-E shown in FIG. **1**. In some embodiments, these facets cause adjacent source/drain

regions **82** of a same FinFET to merge, as illustrated by FIG. **11E**. In other embodiments, adjacent source/drain regions **82** remain separated after the epitaxy process is completed, as illustrated by FIG. **11F**. In the embodiments illustrated in FIGS. **11E** and **11F**, gate spacers **86** are formed covering a portion of the sidewalls of the fins **52** that extend above the STI regions **56** in the transistor region **50X**, thereby blocking the epitaxial growth. In some other embodiments, the spacer etch used to form the gate spacers **86** may be adjusted to remove the spacer material to allow the epitaxially grown region to extend to the surface of the STI region **56** in the transistor region **50X**.

(46) In FIGS. **12A-12D**, a first interlayer dielectric (ILD) **88** is deposited over the structure illustrated in FIGS. **11A-11D**, in accordance with some embodiments. The first ILD **88** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **87** is disposed between the first ILD **88** and the masks **74**, the gate spacers **86**, and the epitaxial source/drain regions **82**. The CESL **87** may comprise a dielectric material, such as silicon nitride, silicon oxide, silicon oxynitride, or the like, having a lower etch rate than the material of the overlying first ILD **88**.

(47) In FIGS. **13** and **14A-14D**, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD **88** with the top surfaces of the dummy gates **72** or the masks **74**, in accordance with some embodiments. FIG. **13** shows a plan view, and FIGS. **14A-14D** show corresponding cross-sectional views. For clarity, the gate seal spacers **80** and the CESL **87** are not shown in FIG. **13**. The planarization process may also remove the masks **74** on the dummy gates **72**, and portions of the gate seal spacers **80** and the gate spacers **86** along sidewalls of the masks **74**. After the planarization process, top surfaces of the dummy gates **72**, the gate seal spacers **80**, the gate spacers **86**, and the first ILD **88** are level. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **88**. In some embodiments, the masks **74** may remain, in which case the planarization process levels the top surface of the first ILD **88** with the top surfaces of the masks **74**.

(48) In FIGS. **15** and **16A-16D**, the dummy gates **72**, and the masks **74** (if present) are removed in one or more etching steps, so that recesses **90** are formed, in accordance with some embodiments. FIG. **15** shows a plan view, and FIGS. **16A-16D** show corresponding cross-sectional views. For clarity, the gate seal spacers **80** and the CESL **87** are not shown in FIG. **15**. Portions of the dummy dielectric layer **60** in the recesses **90** may also be removed. In some embodiments, only the dummy gates **72** are removed and the dummy dielectric layer **60** remains and is exposed by the recesses **90**. In some embodiments, the dummy dielectric layer **60** is removed from recesses **90** in a first region of a die (e.g., a core logic region) and remains in recesses **90** in a second region of the die (e.g., an input/output region). In some embodiments, the dummy gates **72** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **72** with little or no etching of the first ILD **88** or the gate spacers **86**. The recesses **90** may expose portions of the STI regions **56**, in some embodiments. Each recess **90** in the transistor region **50X** exposes and/or overlies a channel region **58** of a respective fin **52**, and each channel region **58** is disposed between neighboring pairs of the epitaxial source/drain regions **82**. During the removal, the dummy dielectric layer **60** may be used as an etch stop layer when the dummy gates **72** are etched. The dummy dielectric layer **60** may then be optionally removed after the removal of the dummy gates **72**.

(49) In FIGS. **17** and **18A-18D**, gate dielectric layers **92** and gate electrode layers **94** are formed in the recesses **90**, forming bottom electrodes **95A** and **95B** in the capacitor region **50C** and forming gate stacks **97** in the transistor region **50X**, in accordance with some embodiments. FIG. **17** shows a plan view, and FIGS. **18A-18D** show corresponding cross-sectional views. For clarity, the gate

seal spacers **80** and the CESL **87** are not shown in FIG. **17**. FIG. **18E** illustrates a detailed view of region **89** of FIG. **18D**. Gate dielectric layers **92** comprise one or more layers deposited in the recesses **90**, such as on the top surfaces and the sidewalls of the fins **52** and on sidewalls of the gate seal spacers **80**/gate spacers **86**. The gate dielectric layers **92** may also be formed on the top surface of the first ILD **88**. In some embodiments, the gate dielectric layers **92** comprise one or more dielectric layers, such as one or more layers of silicon oxide, silicon nitride, metal oxide, metal silicate, or the like. For example, in some embodiments, the gate dielectric layers **92** include an interfacial layer of silicon oxide formed by thermal or chemical oxidation and an overlying high-k dielectric material, such as a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The gate dielectric layers **92** may include a dielectric layer having a k value greater than about 7.0. The formation methods of the gate dielectric layers **92** may include Molecular-Beam Deposition (MBD), ALD, PECVD, and the like. For embodiments in which portions of the dummy dielectric layer **60** remains in the recesses **90**, the gate dielectric layers **92** may include a material of the dummy dielectric layer **60** (e.g., silicon oxide or the like).

(50) The gate electrode layers **94** are deposited over the gate dielectric layers **92**, respectively, and fill the remaining portions of the recesses **90**. The gate electrode layers **94** may include a metal-containing material such as titanium nitride, titanium oxide, tantalum nitride, tantalum carbide, cobalt, ruthenium, aluminum, tungsten, combinations thereof, or multi-layers thereof. For example, although a single gate electrode layer **94** is illustrated in FIGS. **18A-18D**, the gate electrode layers **94** may comprise any number of liner layers **94A**, any number of work function tuning layers **94B**, and a fill material **94C** as illustrated by FIG. **18E**. After the filling of the recesses **90**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layers **92** and the material of the gate electrode layers **94**, which excess portions are over the top surface of the ILD **88**.

(51) The remaining portions of material of the gate electrode layers **94** and the gate dielectric layers **92** in the transistor region **50X** form gate stacks **97** of the resulting FinFETs. The gate electrode layers **94** and the gate dielectric layers **92** may also be collectively referred to as a “replacement gate” or a “gate structure.” The gate stacks **97** may extend along sidewalls of a channel region **58** of the fins **52**.

(52) The remaining portions of material of the gate electrode layers **94** in the capacitor region **50C** form the first bottom electrode **95A** and the second bottom electrode **95B** of the capacitor structure **120**, in accordance with some embodiments. In this manner, the bottom electrodes **95A-B** of a capacitor structure may be formed simultaneously with the gate stacks **97** of the FinFET structure **119**.

(53) By forming electrodes of a capacitor structure along with the gate stacks of transistors, additional electrodes of a capacitor structure can be formed without the use of additional masks or other additional process steps. In some embodiments, one of the bottom electrodes **95A-B** may be connected to a “high” terminal (e.g., a high voltage) and the other of the bottom electrodes **95A-B** may be connected to a “low” terminal (e.g., a low voltage). In some embodiments, both the “high” and the “low” bottom electrodes are formed using the same processes, such as both formed using processes for n-type devices or both formed using processes for p-type devices. In other embodiments, the “high” bottom electrode is formed using processes for n-type devices and the “low” bottom electrode is formed using processes for p-type devices. In other embodiments, the “high” bottom electrode is formed using processes for p-type devices and the “low” bottom electrode is formed using processes for n-type devices.

(54) The capacitance between the first bottom electrode **95A** and the second bottom electrode **95B** (shown schematically in FIG. **18B** and in some subsequent figures) adds to the overall capacitance of the capacitor structure **120**. In some cases, the bottom electrodes **95A-B** may be considered a parallel lateral capacitor that increases the capacitance of the capacitor structure **120**. In some

embodiments, the bottom electrodes **95A-B** may have a rectangular shape with a length **L1** that is in the range of about 100 nm to about 10,000 nm or a width **W1** that is in the range of about 10 nm to about 100 nm, though other dimensions or shapes are possible. In some embodiments, the bottom electrodes **95A-B** may be separated by a distance **D1** that is in the range of about 10 nm to about 1000 nm, though other distances are possible. In some cases, the dimensions or distances may be chosen based on an operating voltage of the device. In some embodiments, the additional capacitance provided to the capacitor structure **120** by the bottom electrodes **95A-B** may be controlled by controlling the length **L1**, width **W1**, and/or separation distance **D1**. For example, increasing the length **L1** of the bottom electrodes **95A-B** or decreasing the separation distance **D1** may increase the additional capacitance. In some embodiments, more than two bottom electrodes **95** may be used, or one or more of the bottom electrodes **95** may comprise two or more separated sections. In this manner, the capacitance of the resulting capacitor structure **120** may also be controlled.

(55) The formation of the gate dielectric layers **92** in the n-type region and the p-type region of the transistor region **50X** may occur simultaneously such that the gate dielectric layers **92** in both type regions (e.g., the n-type region and the p-type region) are formed from the same materials using the same processes, and the formation of the gate electrode layers **94** may occur simultaneously such that the gate electrode layers **94** in both type regions are formed from the same materials using the same processes. Accordingly, the gate dielectric layers **92** and the gate electrode layers **94** in the capacitor region **50C** may be formed from the same respective materials as the gate dielectric layers **92** and the gate electrode layers **94** in the transistor region **50X**.

(56) In some embodiments, the gate dielectric layers **92** in the n-type region of the transistor region **50X** may be formed by processes distinct from the p-type region of the transistor region **50X**. In this manner, the gate dielectric layers **92** and/or the gate electrode layers **94** may be different materials in each type region, which may be formed using distinct processes in each type region. In some embodiments, the gate dielectric layers **92** and the gate electrode layers **94** in the capacitor region **50C** may be formed using the processes and materials used for the n-type region of the transistor region **50X** or formed using the processes and materials used for the p-type region of the transistor region **50X**. For example, both of the bottom electrodes **95A-B** may be formed using the same processes and materials, which may correspond to the processes and materials of the n-type region or to the processes and materials of the p-type region. In some embodiments, one of the bottom electrodes **95A-B** is formed using the processes and materials used for the n-type region and the other of the bottom electrodes **95A-B** is formed using the processes and materials used for the p-type region. Various masking steps may be used to mask and expose appropriate regions when using distinct processes.

(57) In FIGS. **19A-19D**, a gate mask **96** is formed over the gate dielectric layers **92** and/or the gate electrode layers **94**, in accordance with some embodiments. The gate mask **96** may be disposed between opposing portions of the gate spacers **86**. In some embodiments, forming the gate mask **96** includes recessing the gate dielectric layers **92** and/or the gate electrode layers **94** so that a recess is formed directly over the gate dielectric layers **92** and/or the gate electrode layers **94** and between opposing portions of gate spacers **86**. A gate mask **96** comprising one or more layers of dielectric material, such as silicon nitride, silicon oxynitride, or the like, is filled in the recess, followed by a planarization process to remove excess portions of the dielectric material extending over the first ILD **88**. The gate mask **96** is optional and may be omitted in some embodiments. In such embodiments, the gate dielectric layers **92** and the gate electrode layers **94** may remain level with top surfaces of the first ILD **88**.

(58) As also illustrated in FIGS. **19A-19D**, a second ILD **108** is deposited over the first ILD **88**. In some embodiments, the second ILD **108** is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD **108** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD and PECVD.

The subsequently formed electrode contacts **111** (FIGS. **20** and **21A-21B**) and gate contacts **110** (FIGS. **21C** and **21D**) penetrate through the second ILD **108** and the gate mask **96** (if present) to contact top surfaces of the recessed gate electrode layers **94**.

(59) In FIGS. **20** and **21A-21D**, electrode contacts **111**, gate contacts **110**, and source/drain contacts **112** are formed through the second ILD **108** and the first ILD **88** in accordance with some embodiments. FIG. **20** illustrates a plan view of the capacitor region **50C**, FIGS. **21A-21B** illustrate cross-sectional views of the capacitor region **50C**, and FIGS. **21C-21D** illustrate cross-sectional views of the transistor region **50X**. The electrode contacts **111** include electrode contacts **111A** connected to the first bottom electrode **95A** and electrode contacts **111B** connected to the second bottom electrode **95B**, as shown in FIGS. **20**, **21A**, and **21B**.

(60) As an example for forming the electrode contacts **111** and the gate contacts **110**, openings for the electrode contacts **111** and the gate contacts **110** may be formed through the second ILD **108** and the gate mask **96** (if present). In some embodiments, openings for the source/drain contacts **112** may also be formed through the first ILD **88** and the second ILD **108**. The openings may be formed using acceptable photolithography and etching techniques. A liner (not shown), such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD **108**. The remaining liner and conductive material form the electrode contacts **111**, the gate contacts **110**, and the source/drain contacts **112** in the openings. An anneal process may be performed to form a silicide at the interface between the epitaxial source/drain regions **82** and the source/drain contacts **112**.

(61) In the capacitor region **50C**, the electrode contacts **111A-B** are physically and electrically coupled to the bottom electrodes **95A-B**. In the transistor region **50X**, the source/drain contacts **112** are physically and electrically coupled to the epitaxial source/drain regions **82**, and the gate contacts **110** are physically and electrically coupled to the gate electrodes **106**. The electrode contacts **111**, the gate contacts **110**, and the source/drain contacts **112** may be formed in different processes, or may be formed in the same process. In some embodiments, the electrode contacts **111** and the gate contacts **110** are formed using the same process. In some embodiments, top surfaces of the second ILD **108**, the electrode contacts **111**, the gate contacts **110**, and/or the source/drain contacts **112** are level. Although shown as being formed in the same cross-sections, it should be appreciated that each of the source/drain contacts **112** and gate contacts **110** may be formed in different cross-sections, which may avoid shorting of the contacts. Although shown as being formed in the same cross-sections, it should be appreciated that electrode contacts **111A** and electrode contacts **111B** may be formed in different cross-sections. In some embodiments, the electrode contacts **111** have a pitch **P1** that is in the range of about 20 nm to about 200 nm. In some embodiments, the electrode contacts **111A** and the electrode contacts **111B** are separated by a distance **D2** that is in the range of about 10 nm to about 100 nm. In other embodiments, the electrode contacts **111** may have a different size, number, or arrangement than shown.

(62) FIGS. **21C** and **21D** show a FinFET structure **119**, in accordance with some embodiments. Subsequent processing of the transistor region **50X** and the FinFET structure **119** is not shown in subsequent figures, though processing steps similar to those described below for the capacitor region **50C** may be performed on the transistor region **50X**, in some embodiments. It should be appreciated that the disclosed FinFET embodiments could also be applied to nanostructure devices such as nanostructure (e.g., nanosheet, nanowire, gate-all-around, or the like) field effect transistors (NSFETs). In an NSFET embodiment, the fins are replaced by nanostructures formed by patterning a stack of alternating layers of channel layers and sacrificial layers. Dummy gate stacks and source/drain regions are formed in a manner similar to the above-described embodiments. After the dummy gate stacks are removed, the sacrificial layers can be partially or fully removed in channel

regions. The replacement gate structures are formed in a manner similar to the above-described embodiments, the replacement gate structures may partially or completely fill openings left by removing the sacrificial layers, and the replacement gate structures may partially or completely surround the channel layers in the channel regions of the NSFET devices. ILDs and contacts to the replacement gate structures and the source/drain regions may be formed in a manner similar to the above-described embodiments. A nanostructure device can be formed as disclosed in U.S. Patent Application Publication No. 2016/0365414, which is incorporated herein by reference in its entirety.

(63) The formation of bottom electrodes **95A-B** and electrode contacts **111A-B** as described herein can allow for increased capacitance of a capacitor structure **120** (FIGS. **26A-27**). FIG. **22A** shows a plan view similar to that shown in FIG. **20** and FIG. **22B** shows a cross-sectional view similar to that shown in FIG. **21B**. As described previously, the capacitance between the bottom electrodes **95A-B** (indicated schematically in FIGS. **22A-22B**) increases the capacitance of the capacitor structure **120**. Additionally, the electrode contacts **111A-B** as described herein can also increase the capacitance of the capacitor structure **120**. For example, as shown schematically in FIGS. **22A-22B**, the capacitance between electrode contacts **111A** and neighboring electrode contacts **111B** can also contribute to the overall capacitance of the capacitor structure **120**. In some cases, the pairs of electrode contacts **111A-B** may be considered parallel lateral capacitors or additional sets of electrodes that increase the capacitance of the capacitor structure **120**. In this manner, the capacitance of a capacitor structure can be increased without increasing the area of the capacitor structure or without forming additional electrodes at the top of the capacitor structure. In some embodiments, the additional capacitance provided by the electrode contacts **111A-B** may be controlled by controlling the spacing (e.g., the separation distance **D2**), the number, or the arrangement of the electrode contacts **111A-B**. For example, increasing the number of pairs of electrode contacts **111A-B** or decreasing the separation distance **D2** may increase the additional capacitance. In some cases, the use of bottom electrodes **95A-B** and electrode contacts **111A-B** as described herein may increase the capacitance of a capacitor structure by as much as about 15%, though larger increases are possible. In some embodiments, the capacitance contributed by the bottom electrodes **95A-B** may be in the range of about 5% to about 10% of the total capacitance, and the capacitance contributed by the electrode contacts **111A-B** may be in the range of about 2.5% to about 5% of the total capacitance. Other capacitances may be possible.

(64) FIGS. **23** through **27** illustrate various views of intermediate steps in the formation of a capacitor structure **120**, in accordance with some embodiments. FIGS. **23** and **24A-24B** show electrodes **116A'** and **116B'** formed in a dielectric layer **114** over the electrode contacts **111A-B**, and FIGS. **25-27** show additional electrodes **116A-B** formed in additional dielectric layers **115** over the dielectric layer **114**. The electrodes **116A'** in the dielectric layer **114** are electrically coupled to the electrode contacts **111A** and the bottom electrode **95A**, and the electrodes **116B'** in the dielectric layer **114** are electrically coupled to the electrode contacts **111B** and the bottom electrode **95B**. The electrodes **116A** in the dielectric layers **115** are electrically coupled to each other and to the electrodes **116A'** by vias **117A**, and electrodes **116B** in the dielectric layers **115** are electrically coupled to each other and to the electrodes **116B'** by vias **117B**. The electrode **116A'** forms a parallel lateral capacitor with the electrode **116B'**, and each electrode **116A** forms a parallel lateral capacitor with a corresponding electrode **116B**. The capacitance of the pair of electrodes **116A'-B'** and each pair of electrodes **116A-B** adds to the overall capacitance of the capacitor structure **120**. In this manner, the bottom electrode **95A**, the electrode contacts **111A**, the electrode **116A'**, and the electrodes **116A** form a first capacitor electrode **121A** of the capacitor structure **120**, and the bottom electrode **95B**, the electrode contacts **111B**, the electrode **116B'**, and the electrodes **116B** form a second capacitor electrode **121B** of the capacitor structure **120**. In some embodiments, one of the capacitor electrodes **121A-B** may be connected to a “high” terminal (e.g., a high voltage) and the other of the capacitor electrodes **121A-B** may be connected to a “low” terminal (e.g., a low

voltage).

(65) In some embodiments, the electrodes **116A'-B'** and/or the electrodes **116A-B** are formed using the same process that forms metallization patterns, conductive lines, or the like of the device, such as those that may be formed over the transistor region **50X**. For example, a first metallization pattern of the device may be formed in the dielectric layer **114**, with subsequent metallization patterns formed in subsequently formed dielectric layers **115**. In this manner, the dielectric layer **114** and/or the dielectric layers **115** may be considered inter-metal dielectric (IMD) layers, in some cases. The electrodes **116A'-B'** and/or the electrodes **116A-B** may be formed using a suitable process, such as a damascene process, a dual damascene process, or another process. In some embodiments, the electrodes **116A'-B'** and/or the electrodes **116A-B** vertically overlap the bottom electrodes **95A-B**, and the electrodes **116A'-B'** and/or the electrodes **116A-B** may have similar sizes or shapes as the bottom electrodes **95A-B**.

(66) Turning to FIGS. **23** and **24A-24B**, electrodes **116A'-B'** are formed in a dielectric layer **114** over the bottom electrodes **95A-B**, in accordance with some embodiments. FIG. **23** shows a plan view, and FIGS. **24A-24B** show corresponding cross-sectional views. The electrode **116A'** is physically and electrically coupled to the electrode contacts **111A** and the electrode **116B'** is physically and electrically coupled to the electrode contacts **111B**. As shown schematically in FIGS. **23** and **24B**, the capacitance between the electrodes **116A'-B'** provides some of the overall capacitance of the capacitor structure **120** in addition to the capacitance between the electrode contacts **111A-B** and between the bottom electrodes **95A-B**. In some embodiments, the electrodes **116A'-B'** may have a different length than the length **L1** (see FIG. **18A**) of the bottom electrodes **95A-B** or may have a different width than the width **W1** (see FIG. **18B**) of the bottom electrodes **95A-B**. For example, the electrodes **116A'-B'** may have a length greater than **L1** or a width greater than **W1**, though other combinations of length and width are possible.

(67) As an example of forming the electrodes **116A'-B'** shown in FIGS. **23** and **24A-24B**, the dielectric layer **114** may be formed over the second ILD **108** and the electrode contacts **111A-B**. The dielectric layer **114** may be similar to the second ILD **108**, and may be formed using similar techniques. Other materials or techniques are possible. An optional etch stop layer (not shown) may be formed between the second ILD **108** and the dielectric layer **114**. Openings corresponding to pattern of the electrodes **116A'-B'** may then be patterned in the dielectric layer **114**, with the openings exposing surfaces of the electrode contacts **111A-B**. A conductive material may be deposited within the openings to form the electrodes **116A'-B'**. The conductive material may be similar to those described for the electrode contacts **111A-B**, the gate contacts **110**, and the source/drain contacts **112** (see FIGS. **21A-21D**), and may be formed in a similar manner. Other conductive materials or deposition techniques are possible. A planarization process may be performed to remove excess conductive material from the dielectric layer **114**. FIGS. **24A-24B** show the electrodes **116A'-B'** as having substantially vertical sidewalls, but the electrodes **116A'-B'** may have sloped sidewalls, curved sidewalls, or another sidewall profile in other embodiments.

(68) Turning to FIGS. **25** and **26A-26B**, additional electrodes **116A-B** are formed over the electrodes **116A'-B'**, in accordance with some embodiments. FIG. **25** shows a plan view, and FIGS. **26A-26B** show corresponding cross-sectional views. The electrodes **116A-B** may include multiple sets of electrodes **116A-B** formed in multiple dielectric layers **115** over the electrodes **116A'-B'**. In some embodiments, the electrodes **116A-B** are physically and electrically coupled to overlying or underlying features by vias **117A-B**. For example, in the bottom-most dielectric layer **115** over the dielectric layer **114**, the electrodes **116A** are physically and electrically coupled to the electrodes **116A'** by one or more vias **117A**, and the electrodes **116B** are physically and electrically coupled to the electrodes **116B'** by one or more vias **117B**. Each subsequent electrode **116A** in a dielectric layer **115** is physically and electrically coupled to an underlying electrode **116A** by one or more vias **117A**, and each subsequent electrode **116B** in a dielectric layer **115** is physically and electrically coupled to an underlying electrode **116B** by one or more vias **117B**. As schematically

shown in FIGS. 25 and 26B, the capacitance between each pair of electrodes **116A-B** provides additional capacitance for the capacitor structure **120**. The electrodes **116A-B** may have similar dimensions as the electrodes **116A'-B'** or may have different dimensions than the electrodes **116A'-B'**. The vias **117A-B** may or may not overlap the bottom electrodes **95A-B**, and may or may not be aligned with any of the electrode contacts **111A-B**. In some embodiments, the vias **117A** may be located on an opposite side of the capacitor structure **120** from the vias **117B** to reduce the risk of voltage breakdown and/or increase the breakdown voltage of the device. Other arrangements are possible.

(69) The electrodes **116A-B** and vias **117A-B** in the dielectric layers **115** may be formed using a suitable process, such as a damascene process, a dual damascene process, or another process. The electrodes **116A-B** and corresponding vias **117A-B** in a dielectric layer **115** may be formed using separate deposition steps or the same deposition step. The electrodes **116A-B** or vias **117A-B** may be formed of similar materials as the electrodes **116A'-B'**, and may be formed using similar techniques. Each dielectric layer **115** may be formed of one or more layers of materials, and may comprise an etch stop layer in some embodiments. The materials of the dielectric layers **115** may comprise materials similar to those described for the dielectric layer **114** or the second ILD **108**, or may comprise different materials.

(70) FIG. 27 shows a three-dimensional view of a capacitor structure **120**, in accordance with some embodiments. The capacitor structure **120** shown in FIG. 27 is similar to the capacitor structure **120** shown in FIGS. 25 and 26A-B. Some features have been omitted from FIG. 27 for clarity. The capacitor structure **120** shown in FIG. 27 includes a first capacitor electrode **121A** comprising a bottom electrode **95A**, multiple electrode contacts **111A**, an electrode **116A'**, and multiple electrodes **116A** connected by multiple vias **117A** and includes a second capacitor electrode **121B** comprising a bottom electrode **95B**, multiple electrode contacts **111B**, an electrode **116B'**, and multiple electrodes **116B** connected by multiple vias **117B**. The capacitor structure **120** shown in FIG. 27 is an example, and in other embodiments a capacitor structure **120** may have a different arrangement or number of features or the features may have different sizes or shapes. For example, the number or arrangement of electrode contacts **111A-B** or vias **117A-B** may be different than shown. As another example, the number of pairs of electrodes **116A-B** may also be different than shown. In other embodiments, a capacitor structure **120** may comprise zero, one, or more than two pairs of electrodes **116A-B** formed in dielectric layers **115** over the electrodes **116A'-B'**. In this manner, the capacitance of the capacitor structure **120** may be controlled by controlling the number of sets of electrodes **116A-B** formed over the electrodes **116A'-B'**.

(71) As stated previously, the capacitor electrodes **121A-B** may have different shapes or arrangements than shown for the capacitor structure **120** of FIGS. 25-27. As an example, FIGS. 28-30 illustrate a capacitor structure **130** having capacitor electrodes **121A-B** with interdigitated "fingers," in accordance with some embodiments. In some cases, the capacitor structure **130** may be considered a Finger Metal-Oxide-Metal (FMOM) capacitor. FIG. 28 illustrates a three-dimensional view of the capacitor structure **130**, FIG. 29 illustrates a plan view through the electrode contacts **111A-B**, and FIG. 30 illustrates a plan view through the topmost electrodes **116A-B**. For clarity, some features are not shown in FIGS. 28-30. The capacitor structure **130** shown in FIGS. 28-30 has a first capacitor electrode **121A** with three fingers and a second capacitor electrode **121B** with four fingers, but capacitor electrodes **121A-B** may have more or fewer fingers or may have the same number of fingers in other embodiments. In some embodiments, the fingers of one level may be oriented in a different direction (e.g., perpendicularly) to the fingers of another level. The fingers of the capacitor electrodes **121A-B** may have a different arrangement or orientation than shown, and all suitable variations are considered within the scope of the present disclosure.

(72) Referring to FIG. 29, the bottom electrodes **95A-B** and the capacitor electrodes **121A-B** of the capacitor structure **130** are shown, in accordance with some embodiments. Similar to the capacitor

structure **120**, the bottom electrodes **95A-B** may be formed simultaneously with gate stacks **97** (not shown). The capacitance between the bottom electrodes **95A-B** and between the electrode contacts **111A-B** increase the overall capacitance of the capacitor structure **130**. The arrangement of electrode contacts **111A-B** shown is an example, and electrode contacts **111A-B** may have a different arrangement than shown. In some embodiments, the fingers of the capacitor electrodes **121A-B** may have a length **L2** that is in the range of about 100 nm to about 10,000 nm or may have a width **W3** that is in the range of about 10 nm to about 100 nm. In some embodiments, the separation distance **D3** between the capacitor electrodes **121A-B** is in the range of about 10 nm to about 1000 nm. Other lengths, widths, or distances are possible.

(73) FIGS. **1** through **26B** illustrate an embodiment in which FinFET structures **119** are formed in the transistor region **50X**, but other transistor structures may be formed in the transistor region **50X** in other embodiments. As an example, FIGS. **31A-31B** show an embodiment in which a capacitor structure **220** is formed in the capacitor region **50C** and a planar transistor **219** is formed in the transistor region **50X**. FIGS. **31A-31B** are cross-sectional views, with FIG. **31A** illustrated along cross-section B-B and FIG. **31B** illustrated along a cross-section similar to cross-section D-D. Some features of the structures shown in FIGS. **31A-31B** are similar to those described for FIGS. **1-26B** and not all details of all features are repeated. The planar transistor **219** may be n-type or p-type, and the capacitor structure **220** may include features formed using n-type processes and/or p-type processes. Similar to the capacitor electrodes **121A-B** of the capacitor structures **120** or **130**, the capacitor structure **220** includes capacitor electrodes **221A-B** that include bottom electrodes **295A-B**, electrode contacts **211A-B**, electrodes **116A'-B'**, and which may include electrodes **116A-B** (not shown). The bottom electrodes **295A-B** and electrode contacts **211A-B** are formed to provide additional capacitance for the capacitor structure **220**. The embodiment shown in FIGS. **31A-31B** is an example, and other capacitor structures **220** or planar transistors **219** are possible.

(74) In some embodiments, the planar transistor **219** includes STI regions **256** formed in a transistor region **50X** of a substrate **50**. The planar transistor **219** includes a gate stack **297**, which includes a gate dielectric layer **292** formed over the substrate **50** and gate electrode layers **294** formed over the gate dielectric layer **292**, in some embodiments. Gate spacers **286** may be formed along sidewalls of the gate stack **297**, and an etch stop layer **287** may be formed over the gate spacers **286**, gate stack **297**, substrate **50**, and STI regions **256**. Source/drain regions **282** may be formed in the substrate **50** using, for example, one or more implantation processes.

(75) In some embodiments, the capacitor structure **220** is formed on a STI region **256** in a capacitor region **50C** of the substrate **50**, but in other embodiments the capacitor structure **220** is formed on the substrate **50**. The bottom electrodes **295A-B** are formed using the same processes that form the gate stacks **297** of the planar transistor **219**, in some embodiments. For example, the gate dielectric layer **292** may be formed in the capacitor region **50C** and the gate electrode layers **294** may be formed on the gate dielectric layer **292** in the capacitor region **50C**. The bottom electrodes **295A-B** of the capacitor structure **220** are formed from the gate electrode layers **294**, similar to the bottom electrodes **95A-B** of the capacitor structure **120** being formed from the gate electrode layers **94** as described previously.

(76) A first ILD **88** and a second ILD **108** may be formed over the etch stop layer **287**. In the capacitor region **50C**, electrode contacts **211A-B** may extend through the second ILD **108** to physically and electrically couple the bottom electrodes **295A-B**. In the transistor region **50X**, source/drain contacts **210** may extend through the first ILD **88** and the second ILD **108** to physically and electrically couple the source/drain regions **282**, and gate contacts **212** may extend through the second ILD **108** to physically and electrically couple the gate stack **297**. The electrode contacts **211A-B**, the source/drain contacts **210**, and/or the gate contacts **212** may be formed using the same process, in some embodiments. Electrodes **116A'-B'** may be formed on the electrode contacts **211A-B** in a dielectric layer **114** in the capacitor region **50C**, and conductive lines **216** may be formed in the dielectric layer **114** in the transistor region **50X**. The electrodes **116A'-B'** may be

formed using the same process that forms the conductive lines **216**, in some embodiments. Subsequently, sets of electrodes **116A-B** and vias **117A-B** (not shown) may be formed on the electrodes **116A'-B'**, similar to the features shown in FIGS. **26A-B**. In this manner, a capacitor structure **220** may be formed in a capacitor region **50C** and a planar transistor **219** may be formed in a transistor region **50X**.

(77) The embodiments described here have some advantages. The techniques described herein allow for increasing the capacitance of a capacitor structure without the use of additional process steps or masks. The techniques include forming bottom electrodes of the capacitor structure in the same device layer as the gate stack of a transistor, and electrode contacts formed in the same device layer as contacts of the transistor. The bottom electrodes and the electrode contacts can act as additional parallel capacitors of the capacitor structure. Accordingly, the capacitance of the capacitor structure may be increased without increasing the area of the capacitor structure or the vertical height of the capacitor structure above the substrate. This can reduce the manufacturing cost of forming a capacitor structure. The techniques described herein may be fully compatible with Back End of Line (BEOL) processing or various process technologies for forming different types of transistors, such as CMOS process technologies, FinFET process technologies, or the like. The capacitance of the capacitor structure may be tuned by controlling the configuration (e.g., size, shape, separation distance, or the like) of the bottom electrodes and/or the electrode contacts.

(78) In accordance with an embodiment of the present disclosure, a device includes a first dielectric layer over a semiconductor substrate; a second dielectric layer over the first dielectric layer; a transistor over a first region of the semiconductor substrate, wherein the transistor includes a gate stack in the first dielectric layer, wherein the gate stack includes a layer of gate dielectric material and a layer of gate electrode material; and a gate contact in the second dielectric layer, wherein the gate contact is connected to the gate stack, wherein the gate contact includes a gate contact material; and a capacitor structure over a second region of the semiconductor substrate, wherein the capacitor structure includes a first electrode and a second electrode in the first dielectric layer, wherein the first electrode and the second electrode include the gate electrode material; first electrode contacts in the second dielectric layer, wherein the first electrode contacts are over the first electrode and connected to the first electrode, wherein the first electrode contacts include the gate contact material; and second electrode contacts in the second dielectric layer, wherein the second electrode contacts are over the second electrode and connected to the second electrode, wherein the second electrode contacts include the gate contact material, wherein each first electrode contact on the first electrode is adjacent to at least one corresponding second electrode contact on the second electrode. In an embodiment, the capacitor structure further includes a third electrode over and connected to the first electrode contacts and a fourth electrode over and connected to the second electrode contacts. In an embodiment, the third electrode has the same shape as the first electrode, and the fourth electrode has the same shape as the second electrode. In an embodiment, the capacitor structure is on an isolation region in the semiconductor substrate. In an embodiment, the gate electrode material is different from the gate contact material. In an embodiment, the capacitor structure further includes gate dielectric material on the first electrode and on the second electrode. In an embodiment, the transistor is a Fin Field-Effect Transistor (FinFET). In an embodiment, the transistor is n-type.

(79) In accordance with an embodiment of the present disclosure, a semiconductor device includes a substrate; a first device over the substrate, wherein the first device includes a gate stack including a gate electrode material; a source/drain region in the substrate adjacent the gate stack; a first isolation region surrounding the gate stack; a gate contact over and contacting the gate stack, wherein the gate contact includes a gate contact material; and a second isolation region surrounding the gate contact; and a second device over the substrate, wherein the second device includes a first parallel capacitor including first electrodes, wherein the first electrodes include the gate electrode material, wherein the first isolation region separates the first electrodes; and a second parallel

capacitor over the first parallel capacitor, wherein the second parallel capacitor includes second electrodes connected to the first electrodes, wherein the second electrodes include the gate contact material, wherein adjacent second electrodes are separated by the second isolation region. In an embodiment, the first electrodes have interdigitated fingers. In an embodiment, each first electrode is connected to respective second electrodes. In an embodiment, the semiconductor device includes a third parallel capacitor over the second parallel capacitor, wherein the third parallel capacitor includes third electrodes connected to the second electrodes, wherein each third electrode is connected to respective second electrodes. In an embodiment, the first device is a planar transistor. In an embodiment, top surfaces of the second isolation region, the second electrodes, and the gate contact are level. In an embodiment, the semiconductor device includes a first spacer along a sidewall of the gate stack and a second spacer along a sidewall of a first electrode, wherein the first spacer and the second spacer are the same material. In an embodiment, the first parallel capacitor and the second parallel capacitor together provide between 7.5% to 15% of the total capacitance of the second device.

(80) In accordance with an embodiment of the present disclosure, a method of forming a capacitor structure includes depositing a dummy gate material over a semiconductor substrate; patterning the dummy gate material to form dummy electrodes; forming spacers along sidewalls of the dummy electrodes; depositing an isolation material over the dummy electrodes and between the dummy electrodes; removing the dummy gate material to form recesses; depositing a gate dielectric material in the recesses; and forming first electrodes of the capacitor structure, which includes depositing a gate electrode material on the gate dielectric material within the recesses. In an embodiment, the semiconductor substrate includes a fin, and wherein the gate dielectric material and the gate electrode material are deposited over the fin to form a gate stack of a transistor structure. In an embodiment, the method includes forming second electrodes on the first electrodes, wherein forming the second electrodes includes depositing a first dielectric material over the first electrodes; patterning first openings in the first dielectric material that expose the first electrodes; and filling the first openings with a first conductive material. In an embodiment, the method includes forming third electrodes on the second electrodes, wherein forming the third electrodes includes depositing a second dielectric material over the second electrodes; patterning second openings in the second dielectric material that expose the second electrodes; and filling the second openings with a second conductive material.

(81) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a first dielectric layer over a semiconductor substrate; a second dielectric layer over the first dielectric layer; a transistor over a first region of the semiconductor substrate, wherein the transistor comprises: a gate stack in the first dielectric layer, wherein the gate stack comprises a layer of gate dielectric material and a layer of gate electrode material; and a gate contact in the second dielectric layer, wherein the gate contact physically contacts a top surface of the gate stack, wherein the gate contact comprises a gate contact material; and a capacitor structure over a second region of the semiconductor substrate, wherein the capacitor structure comprises: a first electrode and a second electrode in the first dielectric layer, wherein the first electrode and the

second electrode comprise the gate electrode material; a plurality of first electrode contacts in the second dielectric layer, wherein the first electrode contacts physically contact a top surface of the first electrode, wherein the first electrode contacts comprise the gate contact material; and a plurality of second electrode contacts in the second dielectric layer, wherein the second electrode contacts physically contact a top surface of the second electrode, wherein the second electrode contacts comprise the gate contact material, wherein each first electrode contact on the first electrode is adjacent to at least one corresponding second electrode contact on the second electrode, wherein a second electrode contact corresponding to a first electrode contact is a second electrode contact that is closest to that first electrode contact.

2. The device of claim 1, wherein the capacitor structure further comprises a third electrode over and connected to the plurality of first electrode contacts and a fourth electrode over and connected to the plurality of second electrode contacts.

3. The device of claim 2, wherein the third electrode has the same shape as the first electrode, and the fourth electrode has the same shape as the second electrode.

4. The device of claim 1, wherein the capacitor structure is on an isolation region in the semiconductor substrate.

5. The device of claim 1, wherein the gate electrode material is different from the gate contact material.

6. The device of claim 1, wherein the capacitor structure further comprises gate dielectric material on the first electrode and on the second electrode.

7. The device of claim 1, wherein the transistor is a Fin Field-Effect Transistor (FinFET).

8. A semiconductor device comprising: a substrate; a first device over the substrate, wherein the first device comprises: a gate stack comprising a gate electrode material; a source/drain region in the substrate adjacent the gate stack; a first isolation region laterally surrounding the gate stack, wherein the first isolation region extends over the substrate and along a sidewall of the gate stack; a gate contact over and physically contacting the gate stack, wherein the gate contact comprises a first conductive material; and a second isolation region laterally surrounding the gate contact; and a second device over the substrate, wherein the second device comprises: a first parallel capacitor comprising first electrodes, wherein the first electrodes comprise the gate electrode material, wherein the first isolation region laterally separates the first electrodes; and a second parallel capacitor on the first parallel capacitor, wherein the second parallel capacitor comprises second electrodes physically connected to the first electrodes, wherein the second electrodes comprise the first conductive material, wherein adjacent second electrodes are laterally separated by the second isolation region, wherein at least one second electrode that is on one first electrode is linearly between two second electrodes that are on another first electrode.

9. The semiconductor device of claim 8, wherein the first electrodes have interdigitated fingers.

10. The semiconductor device of claim 8, wherein each first electrode is connected to a respective plurality of second electrodes.

11. The semiconductor device of claim 8 further comprising a third parallel capacitor over the second parallel capacitor, wherein the third parallel capacitor comprises third electrodes connected to the second electrodes, wherein each third electrode is connected to a respective plurality of second electrodes.

12. The semiconductor device of claim 8, wherein the first device is a planar transistor.

13. The semiconductor device of claim 8, wherein top surfaces of the second isolation region, the second electrodes, and the gate contact are level.

14. The semiconductor device of claim 8 further comprising a first spacer along a sidewall of the gate stack and a second spacer along a sidewall of the first electrode, wherein the first spacer and the second spacer are the same material.

15. The semiconductor device of claim 8, wherein the first parallel capacitor and the second parallel capacitor together provide between 7.5% to 15% of the total capacitance of the second

device.

16. A method of forming a capacitor structure, the method comprising: depositing a dummy gate material over a semiconductor substrate; patterning the dummy gate material to form dummy electrodes; forming spacers along sidewalls of the dummy electrodes; depositing an isolation material over the dummy electrodes and between the dummy electrodes; removing the dummy gate material to form recesses; depositing a gate dielectric material in the recesses; forming a first electrode and a second electrode of the capacitor structure, comprising depositing a gate electrode material on the gate dielectric material within the recesses; and forming third electrodes on the first electrode and fourth electrodes on the second electrode, comprising depositing a first conductive material directly on the first electrode and the second electrode, wherein each third electrode on the first electrode is laterally adjacent to at least one fourth electrode on the second electrode.

17. The method of claim 16, wherein the semiconductor substrate comprises a fin, and wherein the gate dielectric material and the gate electrode material are deposited over the fin to form a gate stack of a transistor structure.

18. The method of claim 16, wherein forming the third electrodes further comprises: depositing a first dielectric material over the first electrode and the second electrode; patterning first openings in the first dielectric material that expose the first electrode and the second electrode; and filling the first openings with the first conductive material.

19. The method of claim 18 further comprising forming a fifth electrode on the third electrodes, wherein forming the fifth electrode comprises: depositing a second dielectric material over the third electrodes; patterning second openings in the second dielectric material that expose each third electrode; and filling the second openings with a second conductive material, wherein the second conductive material extends continuously over the third electrodes.

20. The device of claim 1, wherein the plurality of first electrode contacts and the plurality of second electrode contacts are free of the first dielectric layer.
